Analog Lab Report Week 5

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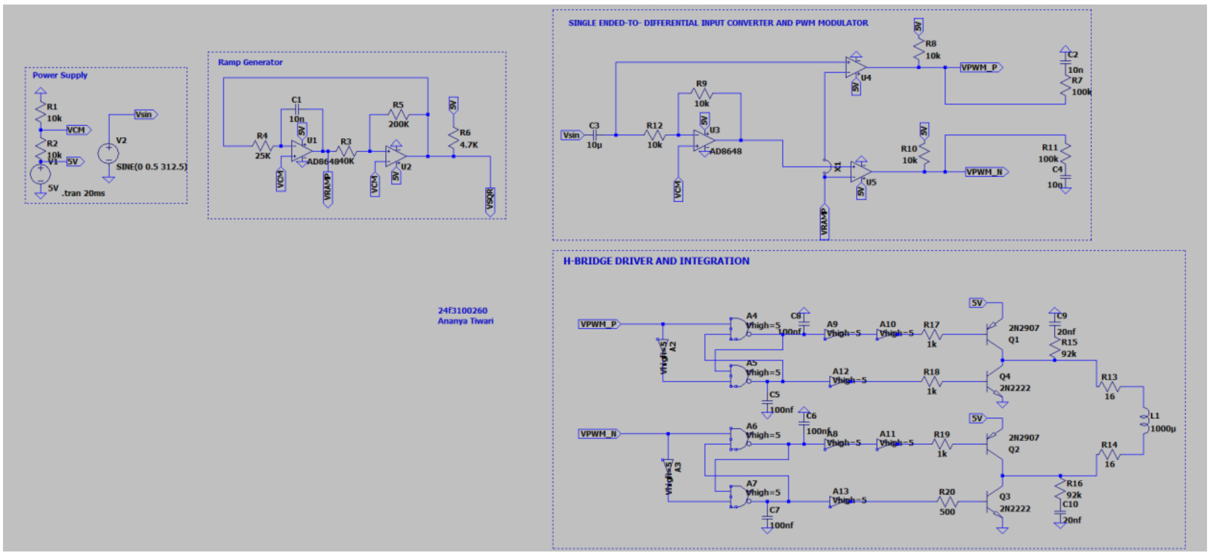
Objective :

This experiment aims to design a half-bridge driver and evaluate a Class-D amplifier utilizing PNP and NPN transistors. We will optimize the base drive circuitry using inverter buffers, starting with simulation in LTSpice before transitioning to hardware implementation and testing.

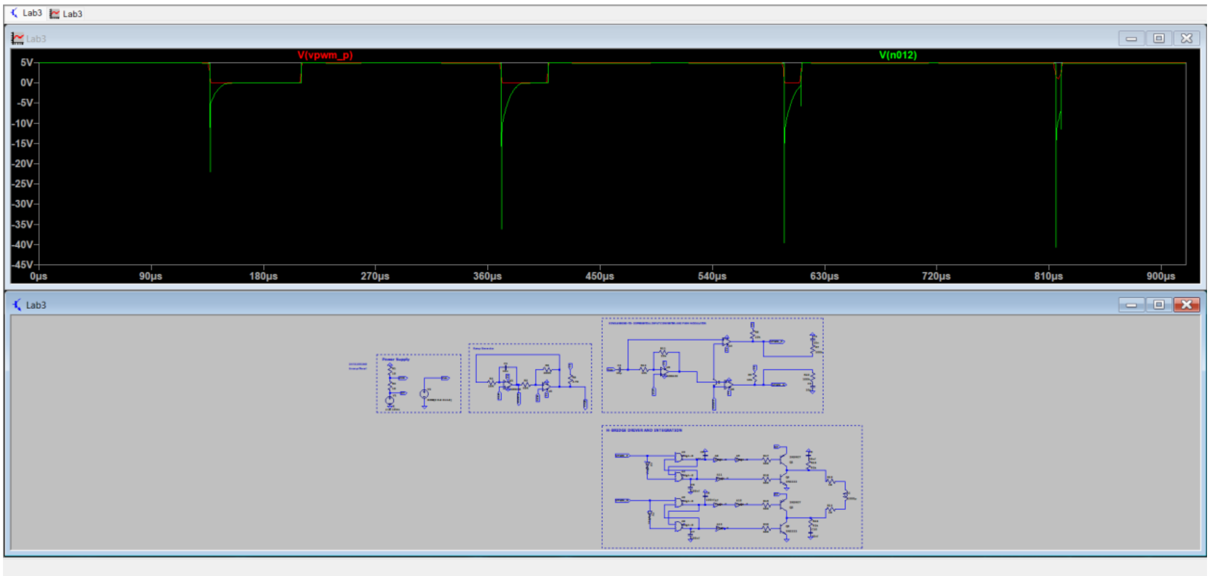
List of Components :

1. Previous Week PWM Modulator
2. 74LSHC00N for NAND Gates
3. CD4096 for inverters
4. NPN (2N2222) and PNP (2N2907)
5. Capacitors and resistors
6. Power Supply (ADALM 1000)
7. Breadboard

LTspice Schematics and Output :

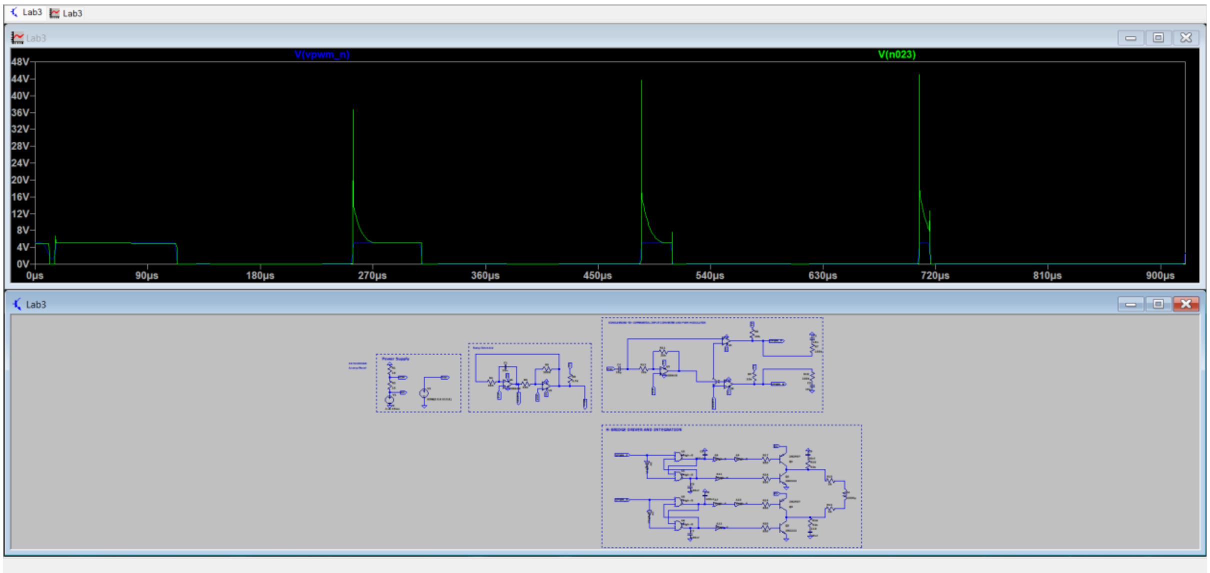


As per the given schematics we were not supposed to put resistor in base but as I was not putting base resistor then transistors were getting short circuited and wasn't working as it is supposed to. After adding them no error is coming but our output signal is more but noisy.



Red square wave: VPWM_P

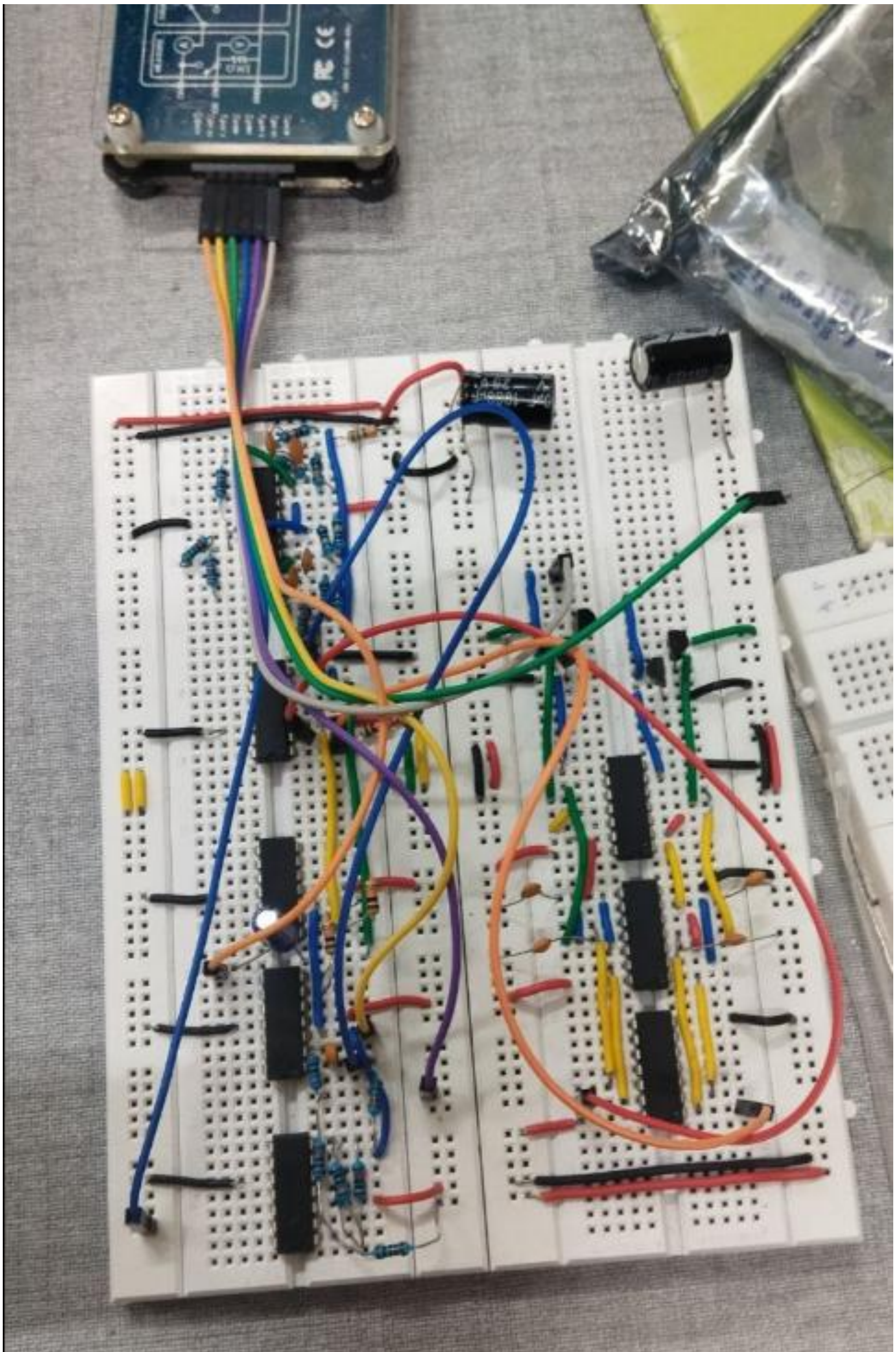
Green square wave: Voutput_P



Blue square wave: VPWP_N

Green square wave : Voutput_P

<u>Physical Circuit :</u>



Physical Circuit Wire Connections :-

Adalmn 1000 wire connections:

1. Blue and green – GND
2. White wire – Channel A (output)
3. Orange wire – Channel B (input)
4. Yellow wire – VDD
5. Purple wire – 2.5 V

1. Orange wire and Red wire – PWM _P and PWM_N wave to the input of Non-Overlap Generator.
2. Blue wire to provide Vsin wave in Week 2 experiment opamp.

Physical Circuit Output :-

Conclusion :-

The experiment was successful, with the physical hardware output validating the simulation models.